

MATRIX G7-S75-02 SERVICE MANUAL



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G7-S75-02

HIP ABDUCTOR

SERVICE MANUAL



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CHAPTER 1 : SERIAL NUMBER LOCATION



Serial Number/EN Label

CHAPTER 2: MAINTENANCE & SAFETY

IMPORTANT SAFETY INFORMATION

It is the sole responsibility of the purchaser of MATRIX products to instruct all individuals, whether they are the end user or supervising personnel on proper usage of the equipment.

It is recommended that all users of MATRIX exercise equipment be informed of the following information prior to its use.

Do not use any equipment in any way other than designed or intended by the manufacturer. It is imperative that MATRIX equipment be used properly to avoid injury.

INSTALLATION

STABLE AND LEVEL SURFACE: MATRIX exercise equipment must be installed on a stable base and properly leveled.

SECURING EQUIPMENT: Manufacturer recommends that all stationary MATRIX strength equipment be secured to the floor to stabilize equipment and eliminate rocking or tipping over. This must be performed by a licensed contractor.

Under no circumstances should you slide equipment across the floor due to risk of tipping. Use proper materials handling techniques and equipment recommended by OSHA. All anchor points must be able to withstand 750 lbf. (3.3 kN) pull-out force.

MAINTENANCE

DO NOT use any equipment that is damaged and or has worn or broken parts. Use only replacement parts supplied by your country's local MATRIX dealer.

MAINTAIN LABELS AND NAMEPLATES: Do not remove labels for any reason. They contain important information. If unreadable or missing, contact your MATRIX dealer for a replacement.

MAINTAIN ALL EQUIPMENT: Preventative maintenance is the key to smooth operating equipment as well as keeping your liability to a minimum. Equipment needs to be inspected at regular intervals.

Ensure that any person(s) making adjustments or performing maintenance or repair of any kind is qualified to do so. MATRIX dealers will provide service and maintenance training at our corporate facility upon request.

ADDITIONAL NOTES

This equipment should only be used in supervised areas where access and control is specifically regulated by the owner. It is up to the owner to determine who is allowed access to this training equipment. The owner should consider a user's: degree of reliability, age, experience, etc.

This training equipment meets industry standards for stability when used for its intended purpose in accordance with the instructions provided by the manufacturer.

This equipment is for indoor use only. This training equipment is a Class S product (designed for use in a commercial environment such as a fitness facility). This training equipment is in compliance with EN ISO 20957-1 and EN 957-2.

CHAPTER 2: MAINTENANCE & SAFETY

PROPER USAGE

1. Do not exceed weight limits of the exercise device.
2. If applicable, set safety stops to appropriate height.
3. If applicable, adjust seat pads, leg pads, foot pads, range of motion adjustment, or any other type of adjustment mechanisms to a comfortable start position. Make certain that the adjusting mechanism is fully engaged to prevent unintentional movement and to avoid injury.
4. Sit on bench (if applicable) and get into appropriate position for exercise.
5. Exercise using no more weight than you can safely lift and control.
6. In a controlled manner, perform exercise.
7. Return weight to its fully-supported start position.

TORQUE VALUES

TORQUE VALUES	
M10 Bolt (Nyloc Nut & Flowdrill)	77Nm / 57lbf -ft
M8 Bolts	25Nm / 18lbf -ft
M8 Plastic	15Nm / 11lbf -ft
M6 Bolts	15Nm / 11lbf -ft
Pad Bolts	10Nm / 7lbf -ft

TOOLS REQUIRED FOR ASSEMBLY&MAINTENANCE(not included)

3MM L-Shaped Allen Wrench



4MM L-Shaped Allen Wrench



5MM L-Shaped Allen Wrench



6MM L-Shaped Allen Wrench



8MM L-Shaped Allen Wrench



10MM L-Shaped Allen Wrench



Phillips Screwdriver



8MM Open-End Wrench



17MM Open-End Wrench



Guide Rod Lubrication



If any items are missing please contact your country's local MATRIX dealer for assistance.



CHAPTER 3: ASSEMBLY INSTRUCTION

1

Do not fully tighten frame connectors until assembly is complete. Vibra-Tite 135 Red Gel or equivalent must be used on all fasteners that are not assembled with Nyloc Nuts.

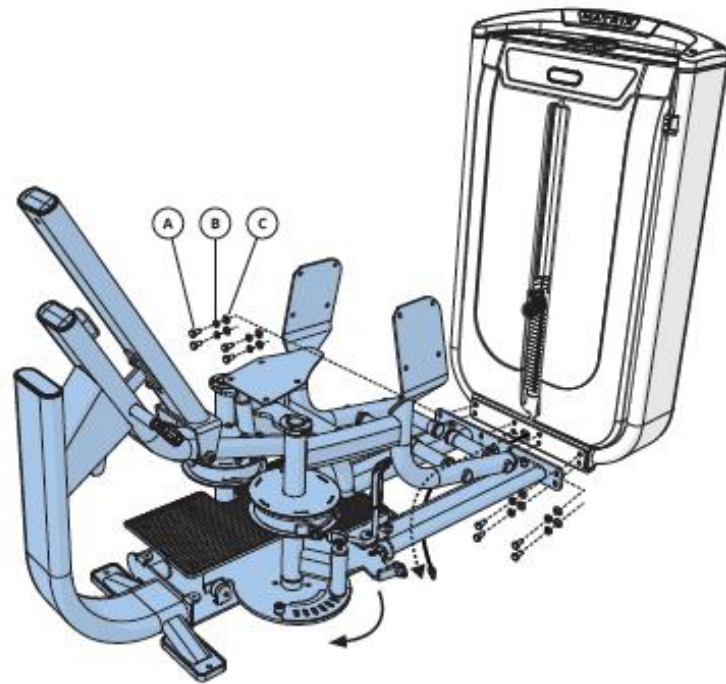
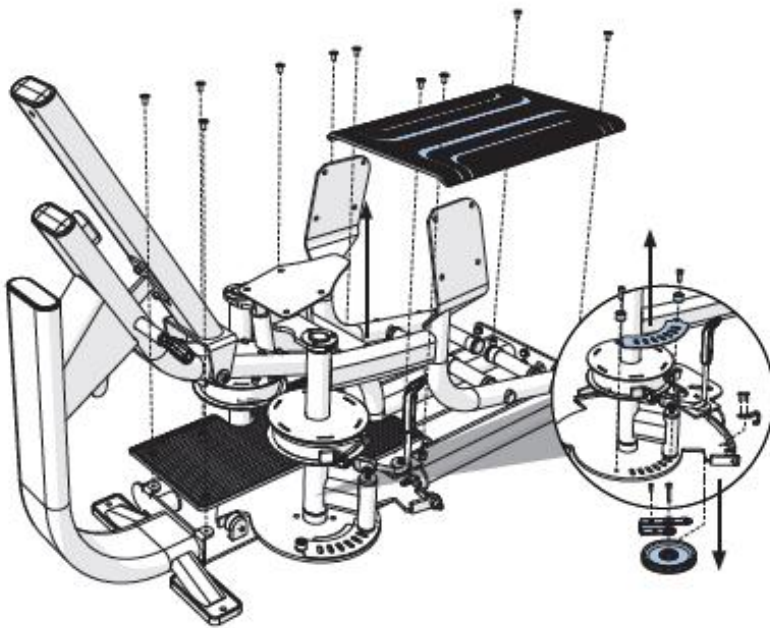
Be sure to remove the insulating sheet when installing the rep counter.



2

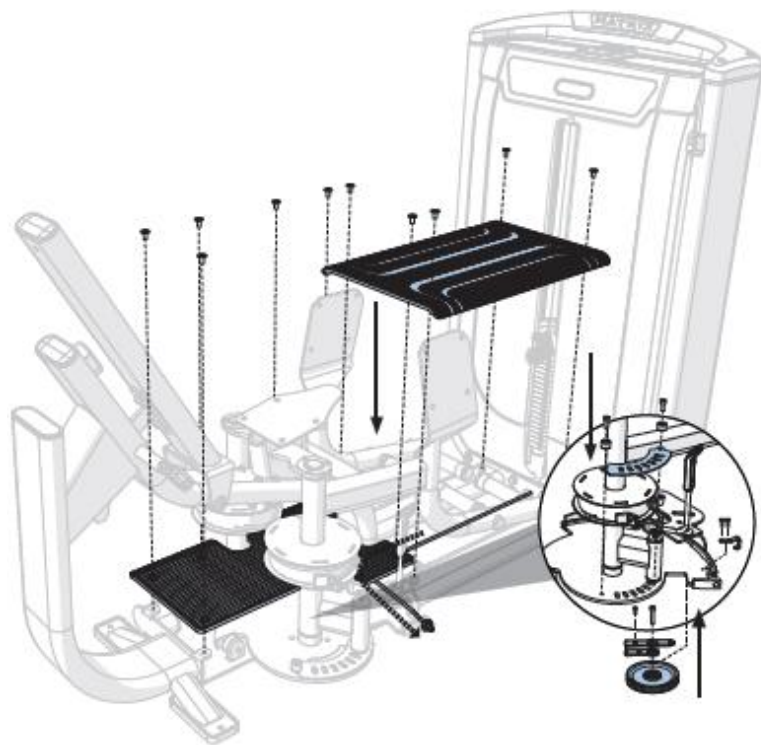
Step 2 Hardware

	Description	Qty
A	Bolt (M10x25L)	8
B	Spring Washer (M10)	8
C	Washer (M10)	8



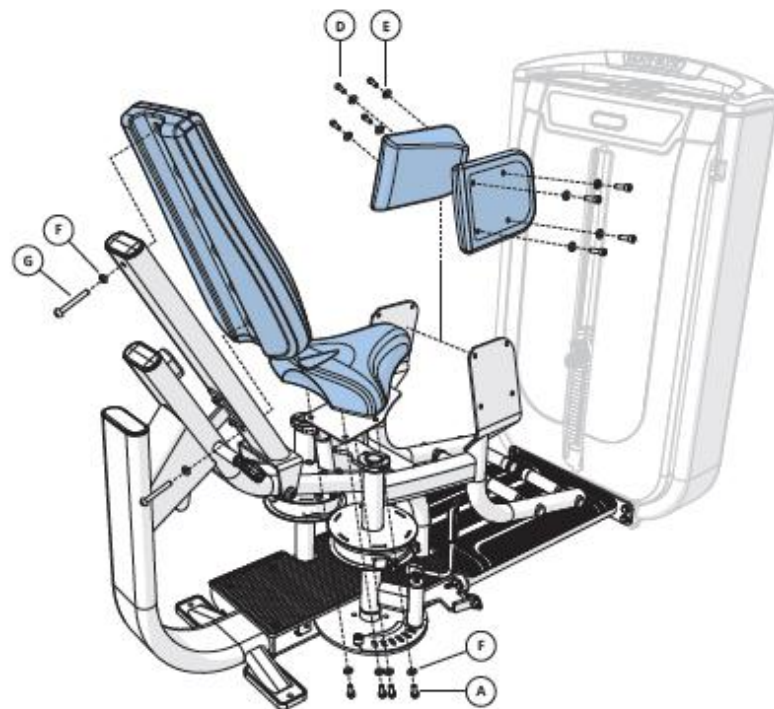
CHAPTER 3: ASSEMBLY INSTRUCTION

3



4

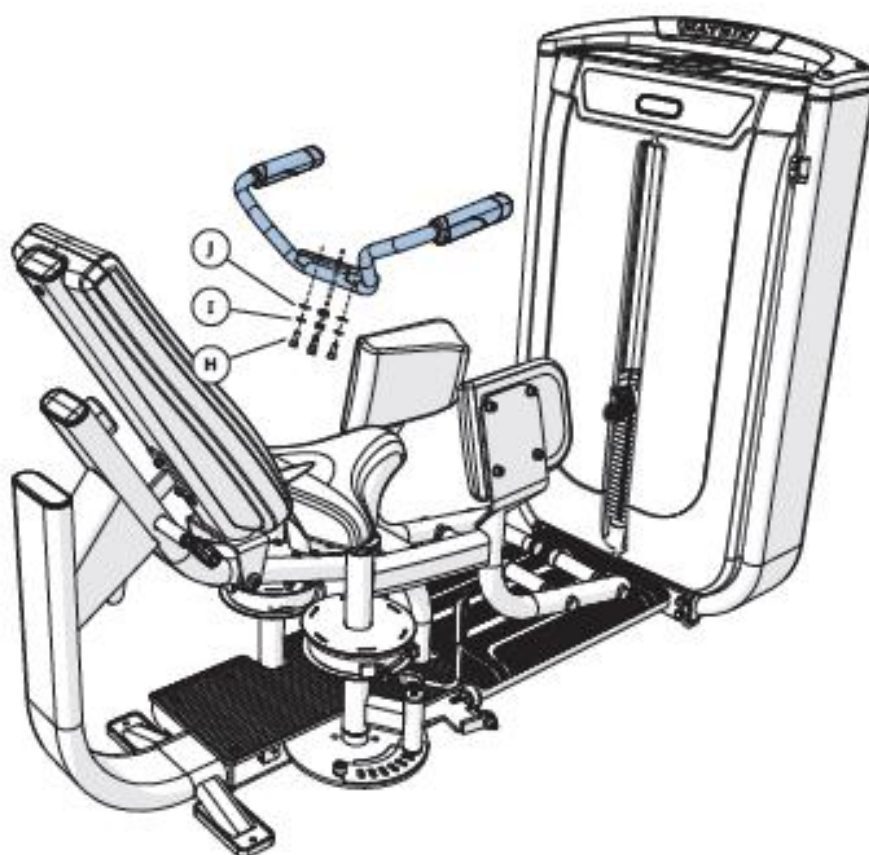
Step 4 Hardware		
Description	Qty	
A Bolt (M10x25L)	4	
D Bolt (M10x30L)	8	
E Washer (Φ10.2x20)	8	
F Washer (Φ10.2x25)	6	
G Bolt (M10x75L)	2	



CHAPTER 3: ASSEMBLY INSTRUCTION

5

Step 5 Hardware		
	Description	Qty
H	Bolt (M8x20L)	4
I	Spring Washer (M8)	4
J	Washer (M8)	4



CHAPTER 4: TROUBLESHOOTING

4.1 THE REPETITION DISPLAY IS BLANK OR IS NOT COUNTING MOVEMENTS.

Possible Causes

- 1) The rep counter battery is dead or disconnected.
- 2) The rep counter is unplugged from the repetition sensors.
- 3) The magnet is missing or positioned incorrectly.
- 4) The repetition sensors are faulty or positioned incorrectly.
- 5) The rep counter is faulty.

Solution

- 1) Remove the 4 screws holding on the aluminum wing and top cover. Check that the battery for the rep counter is firmly seated and has a charge. Replace the battery if needed.
- 2) Check the connection of the wiring coming from the repetition sensors to the rep counter (Figure A).
- 3) Check that the magnet is present and in a position to be read by the repetition sensors (Figure B). The magnet should be slightly off the center of the sensors.
- 4) If the rep counter displays a value, but is not counting repetitions, check the position of the sensors. The sensors should be positioned depending on the model according to the chart on the following page. If the rep sensors are positioned correctly and still not reading, replace the repetition sensors (Figure C).(1000096531)
- 5) If the repetition sensors and battery have been replaced and the issue is still present, replace the rep counter. (1000096514)

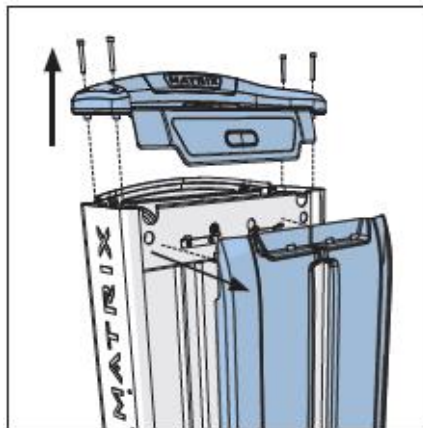


FIGURE A



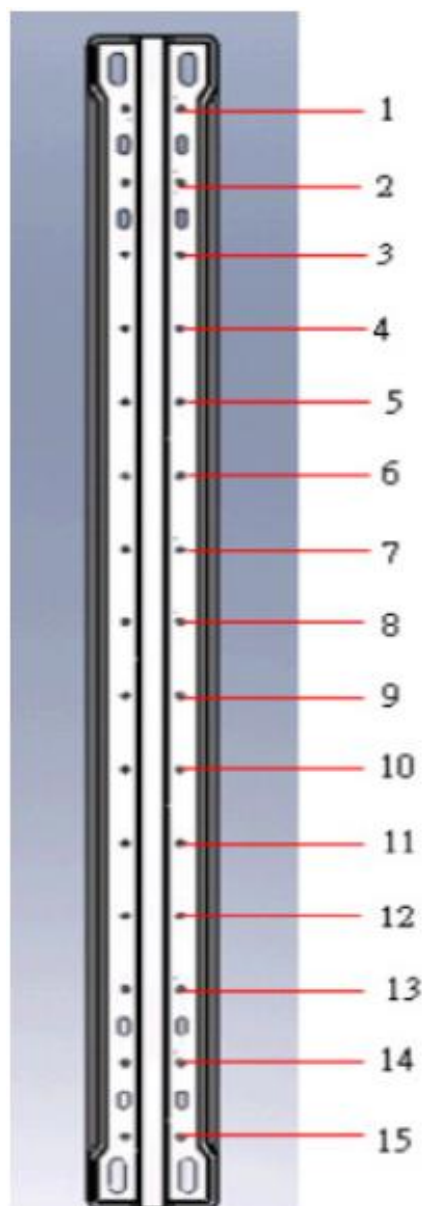
FIGURE B



FIGURE C

CHAPTER 4: TROUBLESHOOTING

4.1 THE REPETITION IS BLANK OR IS NOT COUNTING MOVEMENTS.



MODEL #	A LOCATION	B LOCATION	C LOCATION
Shoulder Press	4	9	12
Chest Press	5	9	13
Lat Pull Down	2	6	12
Seated Row	2	7	13
Leg Extension	7	10	13
Seated Leg Curl	7	11	15
Arm Curl	12	13	15
Tricep Press (Seated Dip)	5	9	12
Abdominal Crunch	10	11	13
Leg Press	10	11	14
Hip Adduction	10	11	13
Hip Abduction	8	9	13
Pec Fly	5	11	15
Back Extension	7	8	13
Prone Leg Curl	8	9	15
Glute	6	11	15
Rotary Torso	11	12	15
Calf Extension	9	11	15
Lateral Raise	7	13	15

CHAPTER 4: TROUBLESHOOTING

4.2 INCREMENTAL WEIGHT PIN IS LOOSE OR FEELS SLOPPY

Possible causes

- 1) The snap ring holding the incremental weight pin in position is damaged or missing.
- 2) The incremental weight pin is damaged.

Solution

- 1) Remove the weight stack covers.
 - a. Remove the 4 screws holding the clear plastic cover over the incremental weight pin and remove the clear plastic cover.
 - b. Check the incremental weight pin for damage, replace if needed. (1000398254)
- 2) Check if the snap ring and washer holding the incremental weight pin to the stack is in place (Figure A & B).
 - a. If this snap ring is missing, install a new snap ring. (1000102910)
 - b. If the snap ring is in place, but not holding the incremental weight pin in place, it may not be properly installed in the groove in the pin.
 - c. If the snap ring is in place and properly installed in the groove, but the pin is still loose, replace the snap ring (the ring may be too stretched out to be effective).

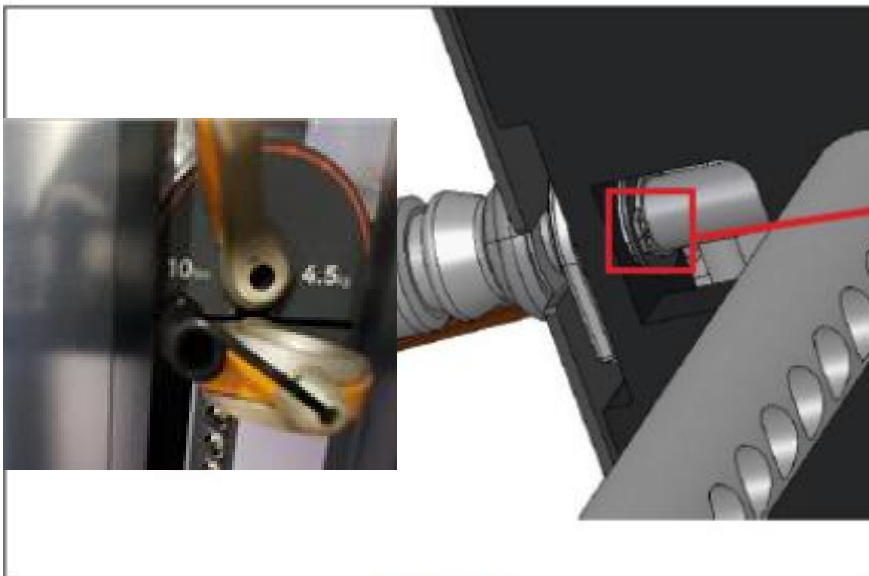


FIGURE A

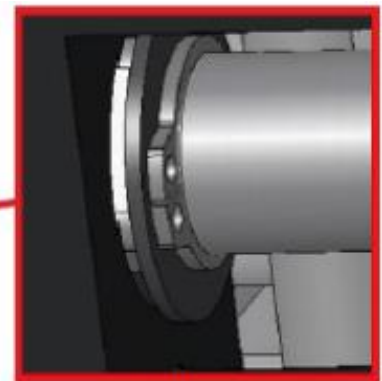


FIGURE B

CHAPTER 4: TROUBLESHOOTING

4.3 WEIGHT STACK PIN WILL NOT GO INTO SOME OR ALL WEIGHT PLATES

Possible Causes

- 1) The weight stack pin is bent.
- 2) The cable is attached too loosely or too tightly.
- 3) The weight plates are not seated correctly at the bottom of the guide rods.
- 4) The top weight plate bushings are damaged.

Solution

- 1) Check the weight stack pin for straightness. Replace if needed (Figure A).(1000398482)
- 2) Check the tightness of the cable. If the cable is too loose or too tight:
 - a. Remove the 4 screws holding down the aluminum wing and top cover. Detach the rep counter wire and set aside the aluminum wing and top cover.
 - b. Loosen the 2 bolts holding the 2 pulleys in place at the top of the weight stack frame (Figure B).

NOTE: A short 8 mm Allen wrench may be needed for some models.

 - c. Rotate the 2 pulley assemblies to remove the slack from the cable, and then re-tighten the 2 pulley bolts.
- 3) Remove the clips holding the weight stack guide rods on the top frame (Figure C).
- a. Lean the guide rods towards the unit frame and shake back and forth so that the weight plates are seated at the bottom of the guide rods without any gap between plates.
- 4) Replace the top weight plate bushings. (1000408503)

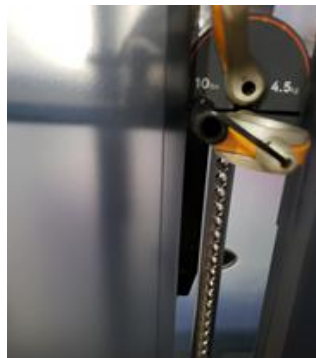


FIGURE A



FIGURE B

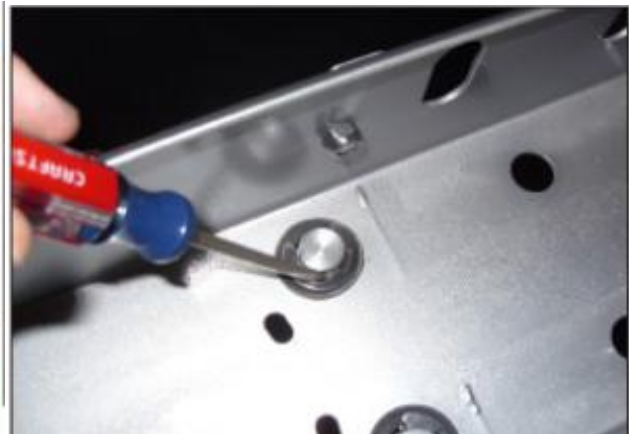


FIGURE C

CHAPTER 4: TROUBLESHOOTING

4.4 THE WEIGHT STACK CABLE IS FALLING OFF A PULLEY OR CAM

Possible Causes

- 1) The cable is too loose or too tight.
- 2) The cable is routed incorrectly.

Solution

- 1) Check the tightness of the cable. Make sure that it is tensioned properly.
- 2) Check the routing of the cable through the weight stack.
 - a. Make sure that the cable is **not** routed around a standoff in the weight stack frame (Figure A).
 - b. Follow the routing of the cable. The cable should be centered on pulleys and cams before being tensioned.

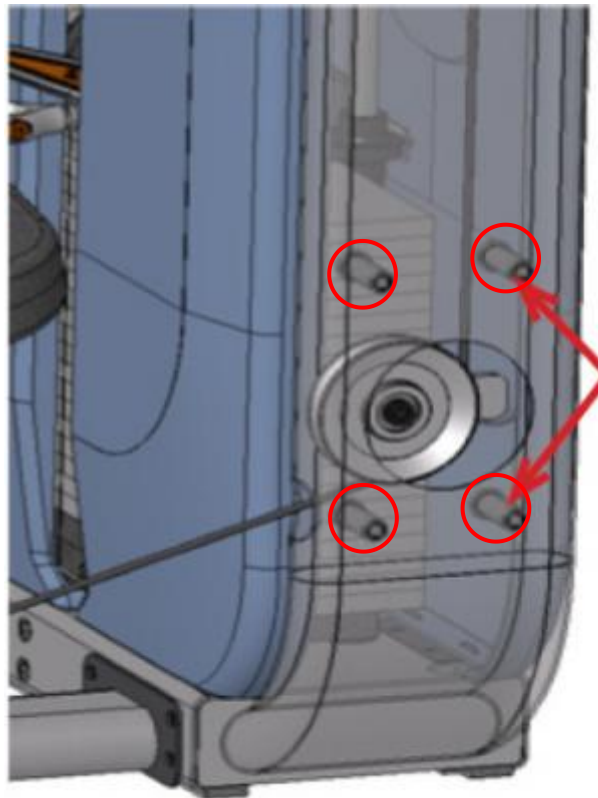


FIGURE A

CHAPTER 4: TROUBLESHOOTING

4.5 REPETITION MAGNET HAS FALLEN OFF OR IS MISSING

Possible Causes

- 1) The repetition counter magnet was not secured tightly enough to the repetition counter bracket.

Solution

- 1) Remove side covers and the 2 screws fixing the magnet frame (Figure A).
- 2) Install a zip tie to securely attach the magnet to the bracket (Figure B).
- 3) When installing the zip tie, make sure that the head of the zip tie is on the opposite side of the bracket from the magnet so that the head does not interfere with the movement of the top weight plate (Figure C).(1000232864)



Figure A

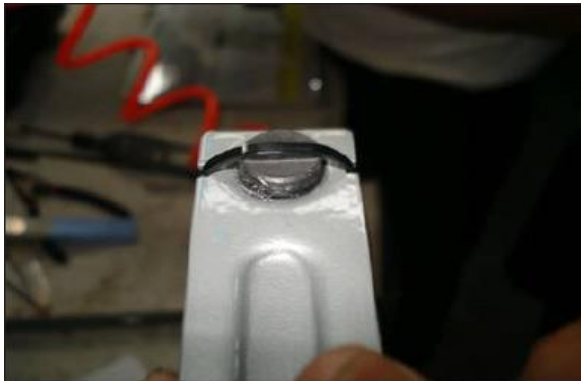


Figure B



Figure C

CHAPTER 4: TROUBLESHOOTING

4.6 ADJUSTABLE HANDLE DOES NOT MOVE EASILY

Possible causes

- 1) The pins are rubbing on the frame creating friction.

Solution

- 1) Remove the affected pin, add grease (Matrix recommends Super Lube brand grease with PTFE) and then re-install the pin.(1000401114/1000423441)



CHAPTER 4: TROUBLESHOOTING

4.7 ARMS ARE UNEVEN OR DON'T MEET IN THE MIDDLE

Possible Causes

- 1) The cables under the seat are not adjusted correctly.

Solution

- 1) Loosen the nuts holding the 2 cables to the cam below the right arm (or seat on the hip units) (Figures A & B).
 - a. Adjust the arms to that they are even and meet in the middle.
 - b. Have someone hold the arms in the correct position (or the arms can be taped together if only one tech is available, just be sure to protect the pads and the paint).
 - c. Once the arms are in place and even, tighten the nuts holding the cables to the cam below the right arm.



FIGURE A



FIGURE B

CHAPTER 5: PARTS REPLACEMENT

5.1 TESTING THE UNIT

Once the unit or replacement part is fully installed and assembled and properly placed on the floor, use the following instructions to test the machine:

NOTE: Not all of the replacement instructions will apply to every model.

- 1) Check that the weight stack pin will slide freely into the weight stack at all positions.
- 2) Check that the incremental weight pin is tight and moves freely.
- 3) Check that the cable is tight and properly routed.
- 4) Check that the seat moves freely and that the seat spring assists the user to move the seat into a higher position.
- 5) Check all adjustment handles for function.
- 6) Check that the arms on the unit are even.
- 7) Check that the repetition counter is active and working properly.

5.2 TOP COVER

- 1) Remove the 4 screws going through the aluminum wing into the top cover of the weight stack (Figure A).
- 2) Disconnect the repetition counter wire and remove the aluminum wing and top cover from the weight stack (Figure B).
- 3) Remove the 4 screws attaching the aluminum wing to the top cover (Figure C).
- 4) Remove the aluminum wing from the top cover (Figure D).
- 5) Reverse Steps 1-4 to install a new top cover. (1000424165/1000397869)



FIGURE A



FIGURE B



FIGURE C



FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.3 REPETITION COUNTER

- 1) Remove the 4 screws going through the aluminum wing into the top cover of the weight stack (Figure A).
- 2) Disconnect the repetition counter wire and remove the aluminum wing and top cover from the weight stack (Figure B).
- 3) Remove the 2 screws attaching the repetition counter to the top cover (Figure C).
- 4) Remove the repetition counter from the top cover (Figure D).
- 5) Reverse Steps 1-4 to install a new repetition counter. (1000096514)



FIGURE A



FIGURE B



FIGURE C



FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.4 SIDE COVER

- 1) Remove the 4 screws going through the aluminum wing into the top cover of the weight stack (Figure A).
- 2) Disconnect the repetition counter wire and remove the aluminum wing and top cover from the weight stack (Figure B).
- 3) Pull up on each weight stack shroud and remove it from the weight stack frame (Figure C).
- 4) Figure D shows the weight stack with all 3 side covers removed.(098985/098988/1000397883)
- 5) Reverse Steps 1-3 to install new side covers.



FIGURE A



FIGURE B



FIGURE C



FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.5 REPETITION SENSOR

- 1) Remove the 4 screws going through the aluminum wing into the top cover of the weight stack (Figure A).
- 2) Disconnect the repetition counter wire and remove the aluminum wing and top cover from the weight stack (Figure B).
- 3) Remove the side covers from the weight stack frame (Figure C).
- 4) Remove the 4 screws holding the repetition sensors to the weight stack frame (Figure D).
- 5) Remove the tie straps holding the repetition sensor wires in place, and remove the repetition sensors.
- 6) Reverse Steps 1-5 to install new repetition sensors.(1000096531)



FIGURE A



FIGURE B



FIGURE C

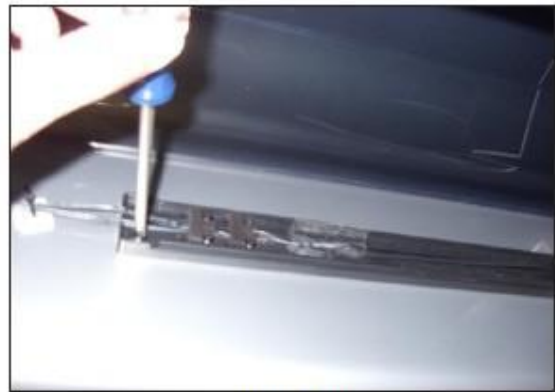


FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.6 INCREMENTAL WEIGHT PIN

- 1) Remove the side covers as outlined in Section 5.4.
- 2) Remove the 4 screws holding on the clear plastic cover over the incremental weight pin and remove it (Figures A & B).
- 3) Remove the 2 screws holding on the reaction block and remove it (Figures C & D).
- 4) Remove the snap ring holding the incremental weight pin to the top weight plate and remove the incremental weight pin (Figures E, F, & G).
- 5) Reverse Steps 1-4 to install a new incremental weight pin.(1000398254)



FIGURE A



FIGURE B



FIGURE C



FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.6 INCREMENTAL WEIGHT PIN

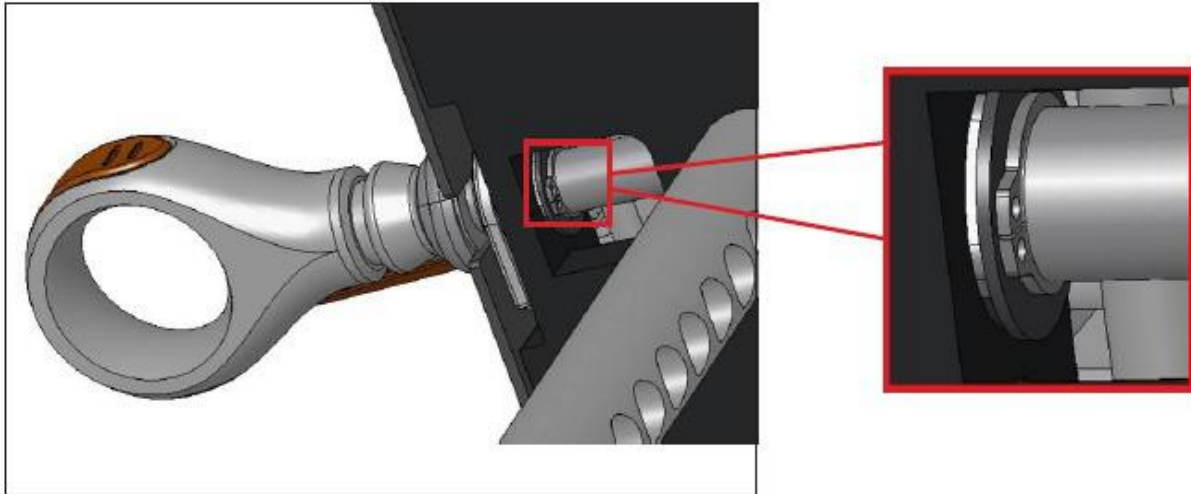


FIGURE E



FIGURE F



FIGURE G

CHAPTER 5: PARTS REPLACEMENT

5.7 WEIGHT PLATE

- 1) Remove the outside cover as outlined in Section 5.2&5.4.
- 2) Remove the 4 screws holding on the clear plastic cover over the incremental weight pin and remove it (Figures A & B).
- 3) Remove the 2 screws holding on the reaction block and remove it (Figures C & D).
- 4) Move the rubber cover on the cable by pulling up on it (Figure E).
- 5) Turn the cable retainer counter clockwise until there is enough slack in the cable to remove it (Figure F & G).
- 6) Remove the incremental weight pin as outlined in Section 5.6.
- 7) Remove the snap ring from the top of the weight stack guide rods (Figure H).
- 8) Lean the guide rod and leave the fixing rack away from the weight stack (Figure I).
- 9) Lean the weight stack towards the frame and remove the weight plates as needed (Figure J).
- 10) Reverse Steps 1-9 to install a new weight plate.(1000092375)



FIGURE A



FIGURE B



FIGURE C



FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.7 WEIGHT PLATE



FIGURE E



FIGURE F



FIGURE G



FIGURE H

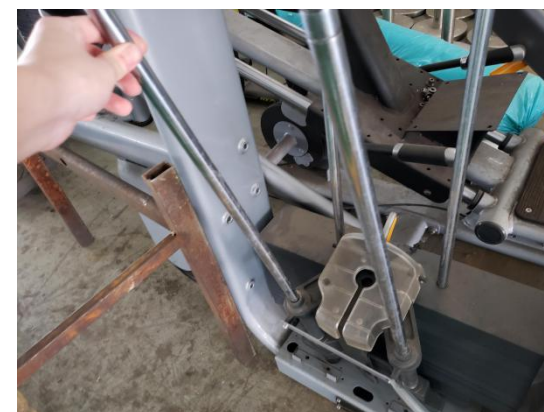


FIGURE I



FIGURE J

CHAPTER 5: PARTS REPLACEMENT

5.8 WEIGHT STUCK PULLEY

- 1) Remove the side covers as outlined in Section 5.4.
- 2) Remove the 4 screws holding on the clear plastic cover over the incremental weight pin and remove it (see Section 5.6 for pictures).
- 3) Move the rubber cover on the cable by pulling up on it (Figure A).
- 4) Turn the cable retainer counter clockwise until there is enough slack in the cable to remove it (Figure B & C).
- 5) Remove the 2 screws fix the weight stack pulley and remove it.(Figure D & F).(1000397609)



FIGURE A



FIGURE B



FIGURE C



FIGURE D

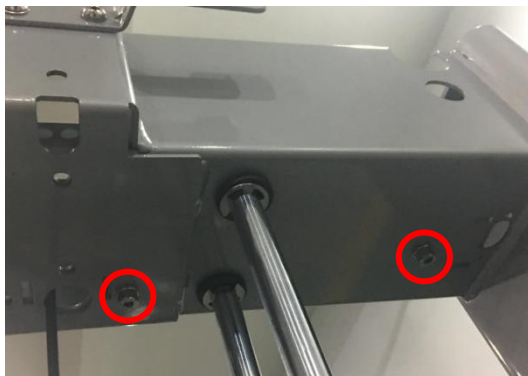


FIGURE F



CHAPTER 5: PARTS REPLACEMENT

5.9 BACK & SEAT & LEG PAD

- 1) Remove the 2 screws to replace the back pad (Figure A&B).
- 2) Remove the 4 screws to replace the seat pad (Figure C).
- 3) Remove 4 screws to replace the arm pad (Figure D).

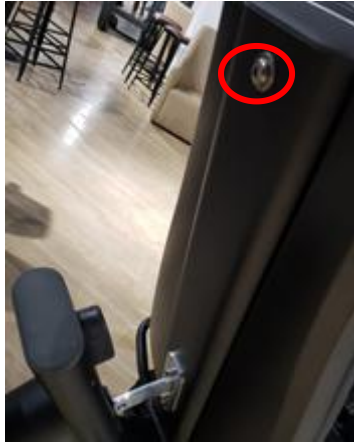


FIGURE A



FIGURE B



FIGURE C



FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.10 ADJUSTABLE HANDLE

- 1) Remove the chain and the screw (Figure A).
- 2) Remove the handle (Figure B).
- 3) Remove the stopper block (Figure C).
- 4) Add grease on the pull pin and reserve step 1-3 to replace(Figure D).(1000401114)

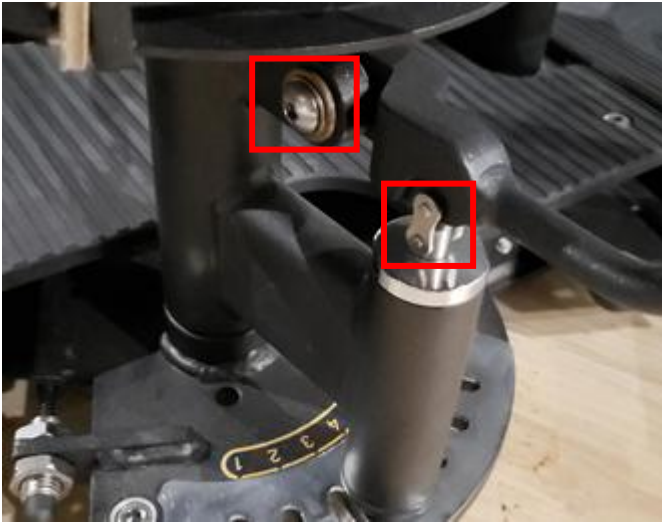


FIGURE A



FIGURE B



FIGURE C



FIGURE D

CHAPTER 5: PARTS REPLACEMENT

5.10 ADJUSTABLE HANDLE - CONTINUE

- 1) Remove the handle (Figure A&B).
- 2) Remove the stopper block (Figure C).
- 3) Remove the pull pin(Figure D).(1000423441)



FIGURE A

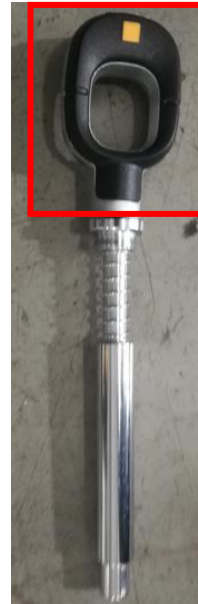


FIGURE B



FIGURE C



FIGURE D

CHAPTER 6: PARTS LIST

Part List	SAP No.	SAP Description
Control board	1000096514	Control Board;Matrix;;H102;S1.60;;GM40;
Sensor	1000096531	Sensor;;;;;24AWG;1200mm;OKI x3+XHS-5Y;
Adjustable pin	1000398254	Adjustment Pin Assembly;Assembly;;weight
Snap ring	1000102910	Snap Ring;External C-Shaped;S-12;;BED;
Adjustable rod	1000398482	Incremental Adjustable Rod-Exreawork;Inc
Fixing axle	1000408503	Fixing Axle;Steel Rope Head;Short;45#;Ch
Magnet	1000232864	Permanent Magnet;;;;;1500gauss;Pull pin;
Pull pin	1000401114	Adjustment Pull pin Set;Adjustable;D;wel
Pull pin	1000423441	Adjustment Pull Pin;back pad;;S45C;Zinc;
Cover (aluminum)	1000424165	Cover;Frame;U;painting;;Black Matte;GM11
Cover (ABS)	1000397869	Cover;;up;ABS/PA757;Deep Grey;740.0x173.
Cover right	098985	Cover;;Right-Front;limpid ABS;;;GM40;
Cover left	098988	Cover;;Left-Front;ABS/PA757;Limpid;;GM40
Cover back	1000397883	Cover;;back;ABS/PA757;Limpid;1104.8x648.
Steel rope	1000398971	Steel Rope-Extrawork;frame;;;Φ6.0;GM115
Steel rope	1000416353	Steel Rope-Extrawork;Link Arm;B;Press Ro
Weight plate	1000092375	Weight Plate Set-Exreawork;-;-;-GM48;
Pulley	1000397609	Pulley;;;NYLON 66+G30%(Anti-fire);Black;

Note : The parts list of Back&Seat&Leg Pad is based on your color; please contact us for more information.

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